

ADIB MAZLOOM

Mechanical & Aerospace Engineer · MBA · US Citizen

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SUMMARY

Mechanical and Aerospace Engineer with full ownership of hardware programs — from ambiguous requirements to detailed CAD, fabricated prototypes, integrated systems, and tested hardware. Experienced designing electromechanical systems, assembling and troubleshooting prototypes on the shop floor, and driving design improvements through hands-on test and failure investigation. Does not hand off after CAD is complete — stays responsible through build, integration, and test. US Citizen.

TECHNICAL SKILLS

Software: SolidWorks · AutoCAD · MATLAB · Python

Design & Analysis: GD&T · Tolerance Analysis · DFM & DFA · Engineering Drawings · FEA (Structural & Thermal)

Fabrication: Additive Manufacturing (FDM, SLA) · Multi-material Prototyping · Hardware Inserts & Post-processing

Integration & Test: Electromechanical Assembly · PCB Integration · Soldering · System-Level Testing · Root Cause Analysis

EXPERIENCE

Lead Mechanical Engineer — Athlete Management Technologies

May 2023 – Present

San Diego, CA

- Sole mechanical engineer owning hardware end-to-end across a product line of charging case variants (multiple form factors) plus separate new product development — from ambiguous requirements through detailed CAD, fabricated prototypes, integrated systems, and validated test, staying responsible at every stage
- Performed structural and thermal FEA to evaluate loads, stiffness, and thermal behavior; turned analysis results into design decisions before cutting any hardware
- Produced detailed GD&T-toleranced drawings and mechanical design packages with DFM/DFA principles embedded from day one — reducing manufacturing time by over 25% on in-house parts and over 90% on outsourced parts
- Assembled and integrated prototype electromechanical systems on the shop floor; diagnosed an inductance mismatch in the pod charger's wireless charging pad and replaced it with a correctly rated component — improving charging efficiency by approximately 30%
- Managed rapid prototyping operation (FDM and SLA) across multiple programs — reduced iteration count from 20 cycles on the 20-unit charger to 9 on the more complex 40-unit charger; personally post-processed, fitted, and tested each iteration
- Collaborated with electrical and software engineers to redesign the pod charger PCB, reducing heat generated by wireless-charging inefficiencies under full-pod load by over 70%

Thermal Design & Test Engineering Intern — Wavious LLC

Jun 2021 – Feb 2022

San Diego, CA

- Developed thermal models and performed first-principles analysis on high-performance electronic hardware; designed and executed physical test setups to validate models against real hardware
- Collected and analyzed test data to drive design changes — improving system thermal performance by 35°C through simulation-informed design iterations

PROJECTS

FireWatch — Autonomous Wildfire Suppression Drone System — Provisional Patent Filed

Apr 2025 – Present

- Designing full mechanical hardware for a two-vehicle autonomous system — airframe structures, propulsion mounts, launcher interface, and payload integration across mothership and suppression drone
- Personally building V1 suppression drone from CAD through physical prototype; provisional patent filed on the tower launcher interface and drone architecture

TrackTechPro — Brake Temperature & DAQ System — Completed

- Designed, wired, and installed a custom Arduino DAQ system integrating thermocouples at all four brake corners and an IMU; built enclosure, routed harness, and mounted on vehicle — hardware and firmware developed independently

EDUCATION

MBA, Cal Poly Pomona · GPA: 3.9 / 4.0 — 2024 – 2025

BS Aerospace Engineering, UC San Diego — 2019 – 2023